

PATHAHEAD · FREE ROADMAP

Become a Cybersecurity Analyst from Scratch

One of the fastest-growing, highest-paid tech fields. Go from zero to job-ready in about five months — with hands-on labs, CTFs, and cert prep.

22

weeks · ~10–12 hrs/wk

5

hands-on projects

4

phases · to job-ready

₹0

free labs & tools

A free PathAhead learning resource · adjust the pace to your life — the order matters more than the dates.

THE ONE RULE

Learn → practice in a lab → document it

Security is a hands-on craft. Everything below is practiced in your own legal lab (TryHackMe, HackTheBox). Only ever test systems you own or are authorized to test.

Legal & ethical use only. Legal & ethical use only. Everything here is for authorized testing, your own lab, or educational platforms. Never attack systems you do not own or have explicit written permission to test — unauthorized access is a crime.

THE ROADMAP

Four phases, step by step

0

Foundations

Weeks 1–5

How computers, networks, and systems actually work.

WEEK 1

Networking

TCP/IP, DNS, HTTP, ports, packets.

WEEK 2

Linux basics

Command line, permissions, bash.

WEEK 3

Windows & systems

Processes, Active Directory basics.

WEEK 4

Security fundamentals

CIA triad, threats, attack surface.

WEEK 5

Cryptography basics

Hashing, encryption, certificates.

1

Offensive Basics

Weeks 6–11

Think like an attacker — ethically and legally.

WEEK 6

Recon & scanning

Nmap, footprinting, enumeration.

WEEK 7

Web vulnerabilities

The OWASP Top 10 hands-on.

WEEK 8

Exploitation basics

Metasploit and common payloads.

WEEK 9

Password attacks

Hashcat, John the Ripper, wordlists.

WEEK 10

Network attacks

Sniffing, MITM, wireless basics.

WEEK 11 · PROJECT ►

Lab writeup

Compromise a vulnerable VM and document it.

2

Defensive & Tooling

Weeks 12–17

Detect, respond, and defend — the blue team.

WEEK 12**Blue team & SIEM**

Log analysis, Splunk/ELK basics.

WEEK 13**Firewalls & IDS/IPS**

Network defense fundamentals.

WEEK 14**Incident response**

Detect, contain, eradicate, recover.

WEEK 15**Malware analysis**

Safe static analysis basics.

WEEK 16**Security scripting**

Python for automation and tooling.

WEEK 17 · PROJECT ►**Home SOC lab**

Set up log monitoring and catch an attack.

3

Practice & Certs

Weeks 18–22

Build proof and prepare for your first cert.

WEEK 18**Guided rooms**

TryHackMe / HackTheBox learning paths.

WEEK 19**Capture The Flag**

CTF challenges to sharpen skills.

WEEK 20**Cert prep**

Study for CompTIA Security+.

WEEK 21**Reporting**

Professional vulnerability reports.

WEEK 22 · CAPSTONE ►**Pentest report**

A full assessment writeup on a lab network.

BUILD A PORTFOLIO

Five projects that get you hired

Projects beat certificates. Do them in order — each builds on the last. Put every one on GitHub with a clear README: problem → approach → result.

Start here

1 • Home Lab Setup

VirtualBox + Kali + a vulnerable VM. Your legal practice range.

Offensive

2 • Vulnerability Writeup

Exploit a TryHackMe box and document every step.

Practice

3 • CTF Walkthroughs

Solve Capture-The-Flag challenges and publish your solutions.

In demand

4 • Home SOC / Monitoring

Collect logs, build alerts, and detect a simulated attack.

Capstone

5 • Full Pentest Report

A professional penetration-test report on your lab network.

ROLES YOU CAN APPLY FOR

Security Analyst • SOC Analyst • Penetration Tester • Security Engineer

TYPICAL STARTING SALARY • INDIA

₹5–12 LPA

LEARN IT

Learning resources

Every item links to the real source (clickable in this PDF).

BOOKS

- [The Web Application Hacker Handbook](#) by Dafydd Stuttard and Marcus Pinto
- [Hacking The Art of Exploitation](#) by Jon Erickson
- [Practical Malware Analysis](#) by Michael Sikorski and Andrew Honig

COURSES

- [Introduction to Cyber Security](#) by NPTEL (free)
- [Google Cybersecurity Certificate](#) by Coursera
- [Cybersecurity Fundamentals](#) by freeCodeCamp (free)
- [Ethical Hacking Essentials](#) by EC-Council

YOUTUBE

- [NetworkChuck](#)
- [John Hammond](#)
- [The Cyber Mentor](#)

PRACTICE

- [TryHackMe](#)
- [Hack The Box](#)
- [PicoCTF](#)

DO THIS TODAY

Your first step (takes 30 minutes)

1

Install VirtualBox + Kali

Free virtualization + the standard security Linux distro.

2

Get a TryHackMe account

tryhackme.com — guided, legal, beginner-friendly hacking labs.

3

Finish one room

Complete the 'Intro to Cyber Security' path today. You've started.

Job-ready checklist: 3–5 solid projects on GitHub (≥ 1 deployed or documented) · can explain your work in an interview · comfortable with the core tools of the field · a portfolio or profile recruiters can find.

PathAhead — India's premium career guidance platform. This roadmap is a free educational resource; timelines are a guide and everyone learns at a different pace. Referenced third-party resources (TryHackMe, HackTheBox, freeCodeCamp, Professor Messer, OWASP) are for learning only and are not affiliated with PathAhead.